

Week2

Be aware: This document does not cover all possible correct answers. If your answer differs, it does not necessarily mean it is incorrect.

Q1 *

10 point

What is the default underlying type of an enum in SystemVerilog if not specified?

- ☐ bit
- ☐ int
- ☐ logic
- ☐ shortint

Answer: int

Q2 *

How many elements are in the following array?

```
bit [7:0] arr [0:3];
```

- ☐ 4 elements, each 8 bits.
- ☐ 8 elements, each 4 bits.
- ☐ 3 elements, each 8 bits.
- ☐ 3 elements, each 7 bits

Answer: 4 elements, each of 8 bits.

Q3 *

10 points

Which statement about SystemVerilog enums is correct?

- ☐ enum types are never synthesizable and should be used only in testbenches.
- ☐ enum types are synthesizable only if all enum labels are explicitly assigned integer values.
- ☐ enum types are synthesizable, and can be used in both design module and testbenches
- ☐ enum types are synthesizable only when declared inside classes.
- ☐ enum types require typedefs to be synthesizable.

Answer:

enum types are synthesizable and can be used in both design modules and testbenches.

Q4) Choose a legal answer for the dynamic array "a" that satisfies the constraint.

(Assume a.size() randomized to be 6)

```
1  class packet;
2      rand bit [7:0] a[];
3
4      constraint c1 {
5          //(assume size got randomized to be 6)
6          a.size() inside {[5:10]};
7          foreach (a[i]) {
8              if (i % 2 == 0)
9                  {a[i][0] == 1;}
10             else
11                 {(a[i] >> 1) << 1 == a[i]; }
12         }
13     }
14 endclass
```

- ☐ {3, 4, 7, 10, 15, 20}
- ☐ {2, 4, 6, 8, 10, 12}
- ☐ {1, 3, 5, 7, 9, 11}
- ☐ {4, 3, 10, 7, 20, 15}

Answer: {3, 4, 7, 10, 15, 20}

Justification:

When we carefully read the constraint, we understand that for each element in the array, it checks the index:

- If the index is even, then the value randomized to be saved in this index should be odd.
- If the index is odd, then the value randomized to be saved in this index should be even.
- $(a[i] \gg 1)$ means apply one shift to the right (losing the least significant bit).
- $(a[i] \gg 1) \ll 1$ shift back to the left one step. (put zero in the least significant bit)

Example : $1101 \gg 1 = (0110) \ll 1 = 1100$

1101 changed to 1100

Applying shift right + shift left forces the value to be even.

Among all the choices provided, we need to search for even values stored in odd indexes and odd values stored in even indexes.

Q5) What is the value of arr[2][3][2:1] after executing the provided code.

```
module tb4;
  reg [4:0] arr [15][10];

  initial begin
    for (int i = 0; i < 15; i = i + 1) begin
      for (int j = 0; j < 10; j = j + 1) begin
        arr[i][j] = i * j;
      end
    end
  end
endmodule
```

- ☐ 2'b00
- ☐ 2'b01
- ☐ 2'b11
- ☐ 2'b10

Answer: 2'b11

Justification:

i = 2, j = 3, arr[i][j] = 6.
6 in decimal = 5'b00110

Q5) What is the displayed value? *

10 points

```
1  module tb4;
2      logic [2:0] arr [15][10];
3
4      initial begin
5          for (int i = 0; i < 15; i = i + 1) begin
6              for (int j = 0; j < 10; j = j + 1) begin
7                  arr[i][j] = i * j;
8              end
9          end
10
11          $display("array = %0b", arr[2][3][3:1]);
12      end
13  endmodule
```

- ☐ array = 011
- ☐ array = 11x
- ☐ array = x11
- ☐ array = 101
- ☐ array = 110

Answer: array = x11

Justification: The elements in the array are of size 3 bits [2:0]. Index 3 is out of range, and therefore it will return x11.

Q6) What is the displayed value? *

1

```
1  module tb2;
2      int da[] = '{1, 2, 3}';
3
4      initial begin
5
6          da = new[5](da);
7
8          for (int i = 0; i < da.size(); i++) begin
9              if (i >= 3) begin
10                 da[i] = i * 2;
11             end
12         end
13
14         $display("da = %0p, d[0] = %0d ", da, da[0]);
15     end
16
17 endmodule
```

- ☐ da = 1 2 3 4 6 , d[0] = 1
- ☐ da = 1 2 4 6 8 , d[0] = 8
- ☐ da = 1 2 3 6 8 , d[0] = 8
- ☐ da = 1 2 3 0 0 , d[0] = 1
- ☐ da = 1 2 3 6 8 , d[0] = 1

Answer: da = 1 2 3 6 8, d[0] = 1

Justification: The old elements will be copied to da, and then for indices 3 and 4, they will hold values that are index*2.

Q7) Which statement about class in SystemVerilog is correct? *

- ☐ A class is a static container like a module
- ☐ A class defines objects that are created dynamically
- ☐ A class can only contain variables, not methods
- ☐ A class instance is automatically created at compile time

Answer: A class defines objects that are created dynamically.

Q8) There are 3 syntax errors in the following module. Mention them all. *

You must find the 3 errors to get the full mark. No partial credits.

```
1  module test_enum_wrong;
2
3      typedef enum logic [2:0] {a=0, b=10, c, d=11} alphabet_t;
4
5      alphabet_t letter;
6
7  initial begin
8      letter = letter.first();
9
10     for(int i=0; i < letter.size(); i++) begin
11
12         $display("letter = %0d", letter);
13
14         letter = letter.next();
15     end
16 end
17 endmodule
```

Your answer

Answer:

1. Enum value out of range 3 bits can represent values from 0 to 7 (line3)
2. c and d will have value 11, which leads to a syntax error. (line3)
3. Incorrect usage of .size() instead of .num() (line10)

Q9) What happens if a constraint is unsolvable during randomization?

- ☐ The simulator will automatically resolve it
- ☐ It returns a random value
- ☐ The randomize function returns 0 (failure)
- ☐ It creates a compile-time error

Answer: The randomize function returns 0 (failure)

Q10) Which SystemVerilog keyword is used to define a new alias for a type?

- ☐ define
- ☐ newtype
- ☐ typedef
- ☐ Typecast

Answer: typedef

Q11) Choose ALL correct answers. *

1

- ☐ Unpacked arrays are not synthesizable and only used in testbenches.
- ☐ Packages can be synthesizable and used in design modules and testbenches
- ☐ Dynamic arrays can be synthesizable if used correctly.
- ☐ none is correct

Answer: Packages can be synthesizable and used in design modules and testbenches.

Q12) ONLY screenshots accepted. No PDFs *

10 points

Constraint two variables such that one is twice the other and their sum is < 50 .

```
class questa_constraint;
  rand bit [7:0] a, b;

  constraint twice {
    // Your code starts here.
  }
endclass

module tb;
  questa_constraint obj = new();

  initial begin
    repeat (5) begin
      void'(obj.randomize());
      $display("a=%0d, b=%0d, sum=%0d", obj.a, obj.b, obj.a + obj.b);
    end
  end
endmodule
```

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```
class questa_constraint;
  rand bit [7:0] a, b;

  constraint twice {
    (a == 2*b) || (b == 2*a);
    (a + b) < 50;
  }
endclass

module tb;
  questa_constraint obj = new();

  initial begin
    repeat (5) begin
      if (obj.randomize()) begin
        $display("a=%0d, b=%0d, sum=%0d", obj.a, obj.b, obj.a + obj.b);
      end
      else begin
        $display("Randomization failed!");
      end
    end
  end
endmodule
```

Answer #1

```
class questa_constraint;
    rand bit [7:0] a, b;

    constraint twice {
        (a == 2*b);
        (a + b) < 50;
    }
endclass

module tb;
    questa_constraint obj = new();

    initial begin
        repeat (5) begin
            if (obj.randomize()) begin
                $display("a=%0d, b=%0d, sum=%0d", obj.a, obj.b, obj.a + obj.b);
            end
            else begin
                $display("Randomization failed!");
            end
        end
    end
endmodule
```

Answer #2